DRAGOON MOUNTAINS GROWING GUIDES

Modern Masterblend Guide

High-Performance Hydroponic Growing in the Dragoon Mountains

Dragoon Mountains Growing Guides

Contents

Figure 1 – The Dragoon Mountains rising above the Sulphur Springs Valley, Cochise County, Arizona.

RELATED GUIDES

This is the purchased-nutrient companion to the Dragoon Mountains library. Pair it with:

- **A — Pepper Growing Guide** — the crop this programme was tuned around
- **B — Bato Bucket Hydroponic Setup** — the recommended build for this programme
- **C — Pepper Hydroponic Guide** — what changes when peppers come off soil
- **D — A Hydroponic Primer** — climate, seasonal calendar, crop list
- **F — Hydroponic Nutrients Guide** — the **locally-sourced** version of this same programme
- **G — pH Up & pH Down Replacements** — homemade alternatives to the bottled acids/bases below

1. The Masterblend System

Masterblend 4-18-38 is a professional-grade, water-soluble fertiliser used by commercial greenhouse growers worldwide. Combined with calcium nitrate and magnesium sulphate, it delivers every essential macro- and micro-nutrient in precise, plant-available form. Mastering the mixing protocol is the foundation of this system.

The 3-Part System

Part	Product	NPK	Primary role	Mix order
A	Masterblend 4-18-38	4-18-38	P, K, Mg + all micronutrients	1st
B	Magnesium sulphate (Epsom salt)	0-0-0	Mg, S — boosts Masterblend Mg	2nd
C	Calcium nitrate 15.5-0-0	15.5-0-0	Ca, N — never mix with A+B concentrate	3rd — always last

Never mix calcium nitrate directly with Masterblend concentrate — they react and precipitate, locking out nutrients. Dissolve each part separately into the reservoir, or keep separate stock solutions.

Mixing Ratios

Use case	Masterblend	Epsom salt	Cal-Nitrate	Target EC	Per volume
Seedlings / cuttings	1.2 g	0.6 g	1.2 g	0.8–1.0	per gallon
Vegetative growth	2.4 g	1.2 g	2.4 g	1.4–1.8	per gallon
Fruiting / flowering	3.0 g	**2.4 g** (peppers) / 1.5 g (tomatoes)	3.0 g	2.0–2.4	per gallon
Heavy fruiting (tomato)	3.6 g	1.8 g	3.6 g	2.4–3.0	per gallon
5-gallon standard (veg)	12 g	6 g	12 g	1.4–1.8	per 5 gal

LOCAL TIP

The pepper Mg rule. Peppers are unusually heavy magnesium users. Once the first fruit is sizing, step Epsom salt up to **2.4 g per gallon** — double the standard rate, and hold there through harvest. This is the single most common Masterblend adjustment in the Dragoons, because soft monsoon rainwater supplies almost no Mg. For tomatoes, the Mg bump is smaller (1.5–1.8 g/gal).

Full Nutrient Profile at Standard Rate (per gallon)

Macro	ppm	Micro	ppm	Micro	ppm
Nitrogen (N)	116	Iron (Fe)	2.4	Zinc (Zn)	0.3
Phosphorus (P)	47	Boron (B)	1.2	Copper (Cu)	0.3
Potassium (K)	188	Manganese (Mn)	1.2	Molybdenum (Mo)	0.06
Calcium (Ca)	113	Chloride (Cl)	12		
Magnesium (Mg)	41	Sulfur (S)	58		

Making Concentrated Stock Solutions

For larger systems or repeated mixing, stock concentrates save time and improve consistency. Keep Stock A and Stock B separate — never combine them as concentrates.

Stock	Contents	Concentration	Dose rate
Stock A	Masterblend 4-18-38 + Epsom salt	1 lb each per gallon of water	1 oz per gallon reservoir
Stock B	Calcium nitrate 15.5-0-0 only	1 lb per gallon of water	1 oz per gallon reservoir

LOCAL TIP

Label stock bottles with the date mixed. Use within two weeks. Store in a cool, dark spot to slow degradation.

RELATED GUIDES

The locally-sourced equivalent of this three-part system — oak-ash potassium, aerated guano for P and N, caliche-vinegar calcium, and epsomite for magnesium — is in **Guide F: Hydroponic Nutrients Guide**.

2. Water Quality & EC Management

SE Arizona well water is among the hardest in the US — high in calcium, magnesium, sodium, and bicarbonates. This directly affects how much Masterblend you add and how stable your pH will be. Knowing your baseline EC is not optional.

Water Quality Targets

Parameter	Ideal starting water	SE AZ well water (typical)	Action if out of range
EC / TDS	< 0.3 mS / < 150 ppm	0.3–0.8 mS / 200–500 ppm	Reduce nutrient dose; consider RO or blend with rainwater
pH	6.5–7.0 (before treatment)	7.5–8.5	Bicarbonate scrub + phosphoric or citric acid
Bicarbonates	< 50 ppm	100–300 ppm	Pre-acidify to neutralise before nutrients
Calcium	< 40 ppm	60–150 ppm	Reduce Cal-Nitrate dose proportionally
Magnesium	< 20 ppm	20–50 ppm	Reduce Epsom dose proportionally

Parameter	Ideal starting water	SE AZ well water (typical)	Action if out of range
Sodium (Na)	< 50 ppm	10–80 ppm	High Na competes with K and Ca — flush if > 80 ppm

Bicarbonate Pre-Treatment Protocol

This is the most important step for Dragoon Mountain growers using well water. Bicarbonates buffer pH upward and fight every pH-down correction. Neutralise them first, then build the nutrient solution on clean water.

Step	Action	What to observe
1	Fill reservoir with source water	Note starting pH and EC
2	Add phosphoric acid (pH Down) drop by drop	Water fizzes — CO_2 releasing as bicarbonates neutralise
3	Wait for fizzing to completely stop	Bicarbonates fully neutralised
4	Adjust pH to 5.8–6.0	Now stable — won't bounce back
5	Add Masterblend, then Epsom, then Cal-Nitrate	In this order, dissolved separately
6	Final check: EC and pH	Adjust if needed, then begin

Best Water Sources for This System

Source	Baseline EC	Suitability	Recommendation
RO filtered water	~0.0	★★★★★ Perfect	Gold standard. Add nutrients from zero.
Monsoon rainwater	< 0.05	★★★★★ Excellent	Collect Jul–Sep. Dark sealed tanks. Nearly as good as RO.
RO / rain 50:50 blend	0.1–0.2	★★★★ Very good	Stretches RO. Consistent and predictable.
Well water (treated)	0.3–0.8	★★★ Usable	Bicarbonate scrub required. Reduce nutrient dose.

Source	Baseline EC	Suitability	Recommendation
Untreated well water	0.3–0.8	★★ Difficult	Inconsistent results. Invest in RO.

LOCAL TIP

A basic RO unit costs $150–$300 and is the single best investment for precision growing in Cochise County. It removes every source-water variable and makes results reproducible.

RELATED GUIDES

For the full toolkit of local acids and alkalis — prickly-pear vinegar, fermented citric, sulphur-burn acid, oak-ash lye, caliche lime — see **Guide G: pH Up & pH Down Replacements**.

3. Modern LED Grow Lights

Modern quantum-board LEDs have transformed indoor growing. They deliver full-spectrum light at high efficiency (2.5–3.0 µmol/J), produce far less heat than HPS, and can be tuned to the exact spectrum each crop needs. In the Dragoons, LEDs also allow year-round growing independent of season or weather.

Key Light Metrics

Term	What it means	Why it matters
PPFD	Photosynthetic Photon Flux Density (µmol/m²/s)	Instantaneous light intensity at canopy. The most important number.
DLI	Daily Light Integral (mol/m²/day)	Total light per day. PPFD × hours ÷ 3600. Varies by crop.
PPE / Efficacy	µmol of light per joule of electricity	Higher = more efficient. Quality LEDs: 2.5–3.0 µmol/J. HPS: ~1.7.
Spectrum	Wavelength blend (nm)	Full spectrum (3000K + 5000K) for veg; add red (660 nm) for fruiting.
Photoperiod	Hours of light per 24-hr cycle	Veg: 18 on / 6 off. Fruiting: 12–16 hrs.

PPFD Targets by Crop & Stage

Crop / stage	PPFD (µmol/m²/s)	DLI (mol/m²/day)	Photoperiod
Seedlings / propagation	100 – 200	5 – 10	18 hrs
Leafy greens (lettuce, kale)	200 – 400	12 – 17	16–18 hrs
Herbs (basil, cilantro)	300 – 500	15 – 20	16–18 hrs
Tomato — vegetative	400 – 600	20 – 25	18 hrs
Tomato — fruiting	600 – 900	25 – 35	14–16 hrs
Pepper — vegetative	400 – 600	18 – 25	18 hrs
Pepper — fruiting	600 – 800	22 – 30	14–16 hrs
Cucumber — fruiting	700 – 1000	30 – 40	14–16 hrs

Recommended LED Models (2025)

Light	Power	Coverage	Efficacy	Best for
Spider Farmer SF-4000	400 W	4×4 ft	2.9 µmol/J	Tomatoes, peppers, high-light crops
Mars Hydro FC-E4800	480 W	4×4 ft	2.85 µmol/J	Heavy fruiting crops
Gavita Pro 1700e LED	645 W	5×5 ft	2.6 µmol/J	Commercial-scale fruiting
Spider Farmer SE-3000	300 W	3×4 ft	2.7 µmol/J	Mixed veg + herb production
HLG 100 V2 (QB)	100 W	2×2 ft	2.7 µmol/J	Seedlings, small herb setups

Dragoon Advantage — Supplemental Lighting

At 4,500–7,500 ft elevation, natural sunlight intensity is 15–20 % higher than at sea level. For greenhouse or semi-outdoor growing, this means natural DLI during fall and spring can reach 40–50 mol/m²/day — more than enough for any crop. LEDs are most valuable in winter (low sun angle) and for extending photoperiod after sunset. Just 2–4 hours per day of supplemental LED lighting in a greenhouse dramatically boosts winter yields. Use shade cloth and LEDs together: shade for midday UV protection, LEDs for early-morning and evening extension.

4. Growing Systems

Three systems stand out for high-performance growing in the Dragoons: Deep Water Culture for leafy crops and smaller fruiting plants, Dutch Bucket for large fruiting crops, and NFT for greens at scale. All pair excellently with Masterblend.

System Comparison

System	Best crops	Complexity	Water use	Masterblend compatibility
Deep Water Culture (DWC)	Tomato, pepper, leafy greens	Low	Very efficient	★★★★★ Ideal
Dutch Bucket (Bato)	Tomato, cucumber, pepper, eggplant	Medium	Efficient	★★★★★ Ideal
NFT (Nutrient Film)	Lettuce, herbs, strawberries	Medium	Most efficient	★★★★★ Ideal
Kratky (passive DWC)	Lettuce, herbs, greens	Very low	Efficient	★★★★ Very good
Ebb & Flow	Most crops	Medium	Moderate	★★★★ Good

RELATED GUIDES

For the full Dutch-bucket build — plumbing, pump sizing, reservoir, daily chores, and the exact Masterblend schedule scaled to it — see **Guide B: Bato Bucket Hydroponic Setup**.

Deep Water Culture (DWC) — Setup

Component	Specification
Reservoir	5–17 gal dark bucket or tote. Insulate or bury for temp control.
Air pump	Minimum 1 W per gallon of reservoir volume. 24/7 operation.
Air stone	Medium or large — maximise dissolved O_2 in root zone.
Net pots	2” or 3” with hydroton or perlite to anchor plants.
Water level	Roots just touching solution at start. Drop 1–2” as roots develop.

Component	Specification
Reservoir temp	65–72 °F — critical. Higher = root rot and low O_2.
Res change	Full change every 7–14 days for fruiting; top off with fresh mix daily.

Dutch Bucket System — Setup

Component	Specification
Buckets	5-gal Dutch or Bato buckets. 1 plant per bucket.
Media	70 % perlite / 30 % hydroton. Rinse thoroughly.
Drip emitters	1 GPH emitter per bucket. Timer: 4–6 cycles per day.
Drain line	Slope ¼” per foot back to reservoir. Single return.
Reservoir	50–100 gal for 8–16 bucket system. Insulate from heat.
Recovery	Recirculating system — collect and reuse drain water.
EC management	Check reservoir EC daily. Top off with half-strength mix as water depletes.

LOCAL TIP

Dutch Bucket with perlite is the number-one choice for tomatoes and peppers in a controlled environment. Excellent root oxygen, perfect drainage, and per-plant EC/pH management if needed.

5. Environmental Control

Nutrients and lights matter, but environment drives the ceiling on your yields. VPD (Vapour Pressure Deficit) is the single most overlooked variable in small-scale growing. Getting it right unlocks faster growth, better nutrient uptake, and dramatically healthier plants.

VPD — Vapour Pressure Deficit

VPD measures the difference between how much moisture the air holds and how much it could hold. Too low = sluggish transpiration, wet leaves, fungal risk. Too high = plants close stomata, nutrient uptake stops, wilting.

Growth stage	Temperature	Humidity	Target VPD (kPa)
Seedlings / clones	72–77 °F (22–25 °C)	70–80 %	0.4 – 0.8
Early vegetative	72–79 °F (22–26 °C)	60–70 %	0.8 – 1.0
Late vegetative	75–82 °F (24–28 °C)	55–65 %	1.0 – 1.2
Flowering / fruiting	68–79 °F (20–26 °C)	45–55 %	1.2 – 1.6
Late fruiting / ripening	65–75 °F (18–24 °C)	40–50 %	1.4 – 1.8

Dragoon Climate Interaction

During the monsoon (July–September), outdoor humidity runs 50–70 %, so VPD drops naturally. Reduce watering frequency and watch for botrytis and powdery mildew. In the dry season (October–June), humidity often sits at 10–30 % and VPD is very high — add a humidifier to your grow space or plants will constantly stress. At elevation, UV is 15–20 % stronger than at sea level, which accelerates leaf-surface drying in a greenhouse with natural light.

Temperature & CO_2

Parameter	Optimal range	Dragoon note
Air temp (lights on)	72–82 °F (22–28 °C)	Summer nights naturally cool — great for fruiting crops that want cold nights
Air temp (lights off)	62–70 °F (17–21 °C)	Dragoon nights often ideal — use outdoor air exchange
Root zone temp	65–72 °F (18–22 °C)	Most critical parameter. Bury or insulate reservoirs. Higher = root rot.
CO_2 (ambient)	400 ppm (natural)	At elevation, ambient CO_2 is slightly lower — sealed rooms benefit from enrichment
CO_2 (enriched)	800–1200 ppm	Boosts photosynthesis 20–30 %. Only effective with high PPFD (600+ µmol/m²/s).
Airflow	Gentle constant movement	Strengthens stems, prevents hot spots, reduces fungal risk during monsoon

6. Nutrient Schedules

Masterblend's simplicity is one of its strengths — the base 2:1:2 ratio (Masterblend : Epsom : Cal-Nitrate) scales linearly across the cycle. You adjust concentration, not ratio, with one exception: **peppers get a larger Mg bump at fruiting.** Charts below are per-gallon working solution.

Tomato — Full Cycle

Week	Stage	MB (g/gal)	Epsom (g/gal)	Cal-N (g/gal)	Target EC	pH
1–2	Germination	—	—	—	0.0	6.0
2–4	Seedling	1.2	0.6	1.2	0.8–1.0	5.8–6.0
4–7	Early veg	2.4	1.2	2.4	1.4–1.8	5.8–6.0
7–10	Late veg / transition	2.8	1.4	2.8	1.8–2.2	5.8–6.2
10–14	First truss flowering	3.0	1.5	3.0	2.0–2.4	5.8–6.2
14–20	Heavy fruiting	3.6	1.8	3.6	2.4–3.0	6.0–6.2
Final 1–2 wks	Flush / finish	0.5	0.3	0.5	0.5–0.8	6.0

Pepper — Full Cycle

Week	Stage	MB (g/gal)	Epsom (g/gal)	Cal-N (g/gal)	Target EC	pH
1–3	Seedling	1.2	0.6	1.2	0.8–1.0	5.8–6.0
3–6	Vegetative	2.4	1.2	2.4	1.4–1.8	5.8–6.0
6–10	Pre-flower	2.8	1.4	2.8	1.8–2.2	5.8–6.2
10+	**Fruiting (Mg bump)**	3.0	**2.4**	3.0	2.2–2.8	6.0–6.2

Pepper fruiting Epsom: 2.4 g/gal, not 1.5. Peppers lift more magnesium through the fruiting phase than any other common hydroponic crop. Hold the Mg bump from first fruit set through harvest. This matches the crop-specific guidance in Section 8.

Leafy Greens & Herbs

Crop	Stage	MB (g/gal)	Epsom (g/gal)	Cal-N (g/gal)	Target EC	pH
Lettuce	All stages	1.2–1.8	0.6–0.9	1.2–1.8	0.8–1.4	5.8–6.2
Basil	Vegetative	2.0	1.0	2.0	1.4–1.8	5.8–6.0

Crop	Stage	MB (g/gal)	Epsom (g/gal)	Cal-N (g/gal)	Target EC	pH
Kale	All stages	2.0–2.4	1.0–1.2	2.0–2.4	1.4–2.0	5.8–6.2
Spinach	All stages	1.6	0.8	1.6	1.0–1.6	6.0–6.5
Cilantro	All stages	1.2–1.6	0.6–0.8	1.2–1.6	0.8–1.4	6.0–6.5

LOCAL TIP

These are starting points. Measure actual reservoir EC and adjust. Tip burn → lower EC. Slow growth with pale leaves → raise EC.

7. pH & EC Monitoring Protocol

Consistent monitoring is what separates good results from great ones. With Masterblend and hard desert water, daily checks for the first two weeks of any new reservoir will reveal your water's behaviour and let you build a predictable maintenance routine.

Daily Monitoring

Frequency	Task	Notes
Morning	pH check	Adjust if outside 5.8–6.2. Note drift direction and amount.
Morning	EC check	Top off with half-strength mix if EC dropped from evaporation.
Morning	Water level	Top off reservoirs. Never let roots dry out.
Evening	Visual plant check	Leaf colour, wilting, pests, root colour.
Weekly	Full reservoir change	Fruiting: every 7–10 days. Greens: every 14 days.
Weekly	Meter calibration	Calibrate pH meter with pH 4.0 and 7.0 buffer.
Weekly	Root inspection	Healthy = bright white. Brown/slippery = problem.
Monthly	System flush	Flush lines and reservoir with plain water between cycles.

pH Adjustment — Precision Protocol

Adjustment	Product	Application	Dragoon local option
pH Down	Phosphoric acid 85 % (GH pH Down)	1 ml per 10 gal, wait 10 min, retest	Fermented prickly-pear vinegar for minor adjustments
pH Up	Potassium hydroxide (GH pH Up)	1 ml per 10 gal, wait 10 min, retest	Dilute oak-ash lye for occasional upward drift
Bicarbonate flush	Citric or phosphoric acid	Pre-treat water until fizzing stops before adding nutrients	Fermented citric acid from local fruit

RELATED GUIDES

Full recipes for each local alternative — strengths, ferment times, safety notes — are in **Guide G: pH Up & pH Down Replacements**.

Understanding EC Drift

EC rising means plants are drinking more water than nutrients — top off with plain RO or rainwater, not a full-strength mix. EC falling means plants are consuming nutrients faster than water — top off with a half-strength Masterblend mix. EC stable but pH rising is the classic bicarbonate effect from hard water — pre-treat next reservoir with acid before adding nutrients. EC stable but pH falling is biological activity or ammonium uptake, common in warm weather; adjust up with pH Up.

8. Crop-Specific Optimisation

Indeterminate Tomato

Parameter	Setting
System	Dutch Bucket (perlite) or DWC
EC	2.0–3.0 mS/cm
pH	5.8–6.2
Light (PPFD)	600–900 µmol/m²/s
DLI	25–35 mol/m²/day
Temp day/night	75–82 °F / 62–68 °F
Photoperiod	14–16 hrs

Parameter	Setting
VPD fruiting	1.2–1.6 kPa
Key tip	Calcium is critical — never skip Cal-Nitrate. Blossom end rot = Ca deficiency or pH lockout.

Chile & Bell Pepper

Parameter	Setting
System	Dutch Bucket or DWC
EC	1.8–2.8 mS/cm
pH	5.8–6.2
Light (PPFD)	600–800 µmol/m²/s
DLI	22–30 mol/m²/day
Temp day/night	72–82 °F / 60–65 °F
Photoperiod	14–16 hrs
VPD fruiting	1.2–1.6 kPa
Key tip	Peppers are heavy Mg users — hold Epsom at **2.4 g/gal** from first fruit set through harvest.

RELATED GUIDES

For heritage Andean peppers (Aji, Rocoto, Manzano) and why the Dragoons' elevation suits them, see **Guide A: Pepper Growing Guide**. For the hydroponic-specific take on all pepper species — including Rocoto's weak points in recirculating systems — see **Guide C: Pepper Hydroponic Guide**.

Lettuce & Leafy Greens

Parameter	Setting
System	NFT or Kratky
EC	0.8–1.6 mS/cm
pH	5.8–6.5
Light (PPFD)	200–400 µmol/m²/s
DLI	12–17 mol/m²/day

Parameter	Setting
Temp	60–72 °F — cool is better
Photoperiod	16–18 hrs
VPD	0.8–1.2 kPa
Key tip	Lower EC produces tender, mild leaves. High EC causes tip burn and bitterness. In summer, grow in the coolest part of your space.

Basil

Parameter	Setting
System	DWC or NFT
EC	1.4–2.0 mS/cm
pH	5.8–6.2
Light (PPFD)	300–500 µmol/m²/s
DLI	15–20 mol/m²/day
Temp	70–85 °F — loves heat
Photoperiod	16–18 hrs
VPD	1.0–1.4 kPa
Key tip	Harvest frequently to prevent bolting. High PPFD produces more essential oils and stronger flavour. An ideal Dragoon monsoon-season crop.

9. Automation & Monitoring

Automation turns a high-maintenance daily task into a manageable weekly check-in. For Dragoon Mountain growers — where temperature swings, monsoon humidity changes, and remote location can make daily attention difficult — even basic automation is a major quality-of-life improvement.

Essential Automation — Priority Order

Device	Function	Cost range	Priority
Digital pH/EC meter	Manual daily monitoring	$30–150	★★★★★ Essential

Device	Function	Cost range	Priority
Digital timer (light cycle)	Automates photoperiod	$15–30	★★★★★ Essential
Inkbird temp controller	Triggers fans/heaters at set temps	$25–50	★★★★★ Essential
Digital hygrometer	Monitors temp + humidity for VPD	$15–25	★★★★ Important
Dosatron / auto-doser	Automatically doses nutrients at set EC	$150–500	★★★ Recommended at scale
Autopilot CO_2 controller	Maintains CO_2 at target ppm	$150–300	★★★ If using CO_2 enrichment
WiFi grow controller (Inkbird, Trolmaster)	Remote monitoring, alerts to phone	$100–400	★★★ Remote-location value
Auto top-off system	Maintains reservoir level	$50–150	★★★ Saves daily labour
EC/pH dosing controller (Bluelab, Atlas)	Continuously monitors and auto-doses	$500–1500	★★ Advanced / commercial

Dragoon-Specific Automation Priorities

Large day/night swings mean reservoir temps can fluctuate significantly — a temp controller with phone alert is high-value here. Rapid humidity swings July–September can push VPD out of range fast; a WiFi hygrometer with alerts is worth it. If you are off-grid or travel frequently, WiFi sensors that alert your phone when pH or temp goes out of range prevent crop losses. Light timers are non-negotiable — consistent photoperiod is critical for fruiting triggers and for preventing premature bolting in greens.

> LOCAL TIP
>
> The Dragoons get 300 + sunny days per year. A modest solar + battery setup (400 W panels + 100 Ah lithium) can power LED lights, pumps, and all monitoring equipment — making this system effectively off-grid.

10. Troubleshooting

Nutrient Deficiency Quick Reference

Symptom	Likely deficiency	Masterblend fix	pH range cause
Yellow between leaf veins — older leaves first	Magnesium (Mg)	Double Epsom dose (2.4 g/gal) for 1–2 weeks	Locked out below 5.5
Yellow between leaf veins — new leaves first	Iron (Fe) or Manganese (Mn)	Check pH — most common cause. Adjust to 5.8–6.0	Locked out above 6.5
Brown / dark tips, necrotic leaf edges	Calcium (Ca) or Potassium (K)	Check Cal-Nitrate was mixed last. Increase to 3.0 g/gal	Locked out below 5.5 or above 7.0
Blossom end rot on tomatoes	Calcium deficiency	Raise Cal-Nitrate to 3.6 g/gal. Check EC isn't too high.	pH out of range or EC too high blocking uptake
Purple/reddish leaf undersides	Phosphorus (P)	Check pH — P locks out below 5.0. Warm to 70 °F+.	Locked out below 5.0 or above 7.5
Overall pale, slow growth	Nitrogen (N) — too low EC	Increase overall dose by 0.5 g/gal	Check EC < 1.0 for fruiting crops
Dark green leaves, stocky but slow	Nitrogen — too high EC	Dilute reservoir. Partial change with fresh water.	EC above 3.5 in most cases

System & Environment Problems

Problem	Cause	Solution
Root rot (brown, slimy roots)	Reservoir temp > 72 °F or insufficient O_2	Insulate reservoir. More airstones. Beneficial bacteria (Hydroguard).
White mineral crust on pots / media	Salt build-up from hard water	Flush with RO or rainwater. Switch primary source. Check EC baseline.
pH unstable — rises rapidly	Bicarbonates in source water	Pre-acidify before nutrients. Switch to RO or rainwater.
Slow growth despite good EC/pH	VPD out of range or root zone too warm	Check temp and humidity. Verify root zone < 72 °F. Check roots.
Powdery mildew on leaves	High humidity + low airflow (common during monsoon)	Increase airflow. Reduce humidity below 60 %. Remove affected leaves.

Problem	Cause	Solution
Light burn — bleached top leaves	LED too close or PPFD too high	Raise light height. Reduce intensity. Check manufacturer guide.
Stretching / etiolation	PPFD too low / not enough light	Lower LED. Increase intensity. Extend photoperiod to 18 hrs.
Algae in reservoir	Light getting into reservoir	Cover all surfaces. Opaque containers only.

Closing

With Masterblend's precision nutrition, modern high-efficacy LED lighting, and controlled-environment techniques, the Dragoon Mountains become an exceptional growing location. Cooler summers than the low desert, abundant solar energy, monsoon rainwater, and long clear fall seasons all work in your favour.

This is the purchased-nutrient programme. The results are reproducible, the numbers are dialled in, and the Mg bump at pepper fruiting is the single most important deviation from the base 2:1:2 ratio. Keep the pepper Epsom at 2.4 g/gal once fruit is sizing, hold pH at 5.8–6.2, and the ceiling on your yields is set by VPD and light — not by nutrition.